

The Theory Atlas

A sheaf-of-theories formalisation of §5.5 with the cocycle condition classifying obstructions

Jawaun Brown Human author and research director

Claude Code (agent) Experiment code, analysis, and manuscript production under direction and review

Date: August 3, 2026 **Status:** two theorems + one worked example (4-bit Boolean world). Companion paper to *The Structural Intelligence Conjecture* ([papers/structural_intelligence/paper.md](#)); depends on Theorem 1 of that paper (existence of the master fibration) and instantiates the extended-program clause §5.5 as two theorems.

Abstract

Extended-program §5.5 of *The Structural Intelligence Conjecture* conjectures that theories can be treated as local charts M_i on contexts U_i with translations $T_{ij} : M_i \rightarrow M_j$, and that where the cocycle $T_{jk} \circ T_{ij} = T_{ik}$ fails the discrepancy is an *informative* obstruction — missing latent, phase transition, or category error. This paper turns that conjecture into two clean theorems.

- **Theorem TA-1 (Cocycle condition is necessary and sufficient for gluing).** A presheaf of theories $\{M_i, T_{ij}\}$ on a context poset extends to a global theory M on the union $\bigcup U_i$ if and only if the cocycle $T_{jk} \circ T_{ij} = T_{ik}$ holds for every triple (i, j, k) with pairwise overlap. Standard sheaf/descent argument, executed in the discrete-poset setting with a target label alphabet.
- **Theorem TA-2 (Cocycle failure classifies obstructions).** If the cocycle fails, the discrepancy $D := T_{ik}^{-1} \circ T_{jk} \circ T_{ij}$ is a permutation on the target alphabet whose rank and edge-support classify the obstruction: rank-zero on every triple = **glue**; rank ≥ 1 with the non-identity transitions supported on a single pairwise overlap = **phase transition / boundary**; rank ≥ 1 with every pairwise transition non-identity = **missing latent**.

The theorems are elementary: TA-1 is the discrete analogue of the Čech-descent equivalence familiar from sheaf theory, executed on a context poset without measure-theoretic overhead; TA-2 reads the same equivalence backwards to make the *shape* of the failure into a diagnostic.

An exact instrument ([experiments/theory_atlas_pair](#)) exhibits both on the 4-bit Boolean world of Instrument 4. Three contexts $U_1 = \{x_0 = 0\}$, $U_2 = \{x_1 = 0\}$, $U_3 = \{x_2 = 0\}$ share the same chart maps into a target $T = \mathbb{Z}/4$; the good family $(T_{12}, T_{23}, T_{13}) = (+1, +1, +2)$ satisfies the cocycle and glues to the global theory $M(x) = (2 \cdot x_2 + x_3) \bmod 4$; the bad family $(+1, +1, +3)$ fails the cocycle with discrepancy shift-by-3 (rank 4, no fixed points), does not glue on 6 of 14 union worlds, and matches the missing-latent signature exactly. A third $(+1, \text{id}, \text{id})$ phase-boundary reference is evaluated in parallel as a control that TA-2's taxonomy separates the two regimes on identical machinery. All four pre-registered gates pass exactly.

1. Setup

We inherit the master object of *The Structural Intelligence Conjecture* §1: a stochastic fibration $(q : X \rightarrow Z, K : Z \rightsquigarrow X)$ on a standard Borel X . Here we do *not* fix a global q ; instead we consider a family of local charts, each carrying its own coarse view.

Contexts. A finite index set A with a *context poset* $\{U_\alpha \subseteq X : \alpha \in A\}$ partially ordered by inclusion. Overlaps $U_{\alpha\beta} := U_\alpha \cap U_\beta$ and triple overlaps $U_{\alpha\beta\gamma} := U_\alpha \cap U_\beta \cap U_\gamma$ are the pairwise and triple intersections of the covering family.

Charts. A *chart* on context U_α is a measurable map $M_\alpha : U_\alpha \rightarrow \mathcal{T}$ into a fixed target alphabet $\mathcal{T}$. In this paper $\mathcal{T}$ is finite;

the extension to a general standard Borel target is the same argument with regular conditional kernels replacing set functions.

Transitions. For every pair of overlapping contexts, a *transition* $T_{\alpha\beta} : \mathcal{T} \rightarrow \mathcal{T}$ —a bijection on $\mathcal{T}$ —encoding how a chart- α label is re-expressed in chart- β 's alphabet on $U_{\alpha\beta}$. Concretely, the transitions are the identifications that would let a chart- α observer translate their measurements to chart- β 's coordinates without loss.

Presheaf of theories. The data $\mathcal{P} := (\{M_\alpha\}, \{T_{\alpha\beta}\})$ is a *presheaf of theories* on the context poset: to every context we assign a chart (a "theory of the local context") and to every overlap we assign a translation.

Gluing. A *global theory* is a measurable map $M : U_\alpha \cup_\alpha U_\alpha \rightarrow \mathcal{T}$ together with a family of bijections $\psi_\alpha : \mathcal{T} \rightarrow \mathcal{T}$ such that $M(x) = \psi_\alpha(M_\alpha(x))$ for all $x \in U_\alpha$ and every α . We say $\mathcal{P}$ *glues* if such a $(M, \{\psi_\alpha\})$ exists.

Cocycle. The presheaf $\mathcal{P}$ satisfies the *cocycle condition* if, for every triple (α, β, γ) with non-empty pairwise overlaps,

$$T_{\beta\gamma} \circ T_{\alpha\beta} = T_{\alpha\gamma} \quad \text{as maps } \mathcal{T} \rightarrow \mathcal{T}.$$

Equivalently, the *cocycle discrepancy*

$$D_{\alpha\beta\gamma} := T_{\alpha\gamma}^{-1} \circ T_{\beta\gamma} \circ T_{\alpha\beta}$$

is the identity permutation on $\mathcal{T}$ for every triple.

The rest of the paper proves that gluing is equivalent to the cocycle condition (Theorem TA-1) and that the failure of the latter classifies the shape of the obstruction (Theorem TA-2).

2. Theorem TA-1: cocycle is necessary and sufficient for gluing

Theorem TA-1 (Cocycle condition is necessary and sufficient for gluing). A presheaf of theories $\mathcal{P} = (\{M_\alpha\}, \{T_{\alpha\beta}\})$ on a context poset extends to a global theory $M : U_\alpha \cup_\alpha U_\alpha \rightarrow \mathcal{T}$ if and only if the cocycle condition $T_{\beta\gamma} \circ T_{\alpha\beta} = T_{\alpha\gamma}$ holds for every triple (α, β, γ) with non-empty pairwise overlaps.

Proof.

(Necessity.) Suppose $\mathcal{P}$ glues via $(M, \{\psi_\alpha\})$. Then for every α, β with $U_{\alpha\beta} \neq \emptyset$ and every $x \in U_{\alpha\beta}$,

$$M(x) = \psi_\alpha(M_\alpha(x)) = \psi_\beta(M_\beta(x)),$$

so $M_\beta(x) = (\psi_\beta^{-1} \circ \psi_\alpha)(M_\alpha(x))$ on the overlap. Because the ψ_α 's are bijections of $\mathcal{T}$, they act pointwise on labels and the identity $M_\beta = (\psi_\beta^{-1} \circ \psi_\alpha) \circ M_\alpha$ on $U_{\alpha\beta}$ forces

$$T_{\alpha\beta} = \psi_\beta^{-1} \circ \psi_\alpha$$

(uniquely on the image of M_α ; extended arbitrarily to all of $\mathcal{T}$, but the composition identities below use only image values). Then

$$\begin{aligned} T_{\beta\gamma} \circ T_{\alpha\beta} &= (\psi_\gamma^{-1} \circ \psi_\beta) \circ (\psi_\beta^{-1} \circ \psi_\alpha) \\ &= \psi_\gamma^{-1} \circ \psi_\alpha \\ &= T_{\alpha\gamma}. \end{aligned}$$

So the cocycle holds on every triple.

(Sufficiency.) Assume the cocycle. Fix any base chart index (WLOG $\alpha = 0$), set $\psi_0 := \text{id}$, and define

$$\psi_\beta := T_{0\beta}^{-1} \quad \text{for every } \beta \neq 0.$$

Define $M(x) := \psi_\beta(M_\beta(x))$ for any β with $x \in U_\beta$. We show that M is single-valued on $U_\beta \cup_\beta U_\beta$.

Take $x \in U_\beta \cap U_\gamma$ and suppose $\beta, \gamma \neq 0$. By the pairwise identification on $U_{\beta\gamma}$,

$$M_\gamma(x) = T_{\beta\gamma}(M_\beta(x)),$$

hence

$$\psi_\gamma(M_\gamma(x)) = T_{0\gamma}^{-1}(T_{\beta\gamma}(M_\beta(x))).$$

Applying the cocycle to the triple $(0, \beta, \gamma)$,

$$T_{\beta\gamma} \circ T_{0\beta} = T_{0\gamma} \implies T_{0\gamma}^{-1} \circ T_{\beta\gamma} = T_{0\beta}^{-1} = \psi_\beta,$$

so $\psi_\gamma(M_\gamma(x)) = \psi_\beta(M_\beta(x))$. The $0 \in \{\beta, \gamma\}$ case is direct because $\psi_0 = \text{id}$ and $T_{0\beta}^{-1} \circ M_\beta = \psi_\beta \circ M_\beta$ on $U_{0\beta}$ by definition. So M is well-defined on the union, and the pair $(M, \{\psi_\alpha\})$ witnesses gluing. $\square$

Corollary (uniqueness up to relabelling of $\mathcal{T}$). *If two global theories $(M, \{\psi_\alpha\})$ and $(M', \{\psi'_\alpha\})$ witness gluing of the same presheaf, then $M' = \pi \circ M$ for a single bijection $\pi : \mathcal{T} \rightarrow \mathcal{T}$ and $\psi'_\alpha = \pi \circ \psi_\alpha$ for every α . Direct from $M'(x) = \psi'_\alpha(M_\alpha(x)) = (\psi'_\alpha \circ \psi_\alpha^{-1})(M(x))$, with the composition $\pi := \psi'_\alpha \circ \psi_\alpha^{-1}$ independent of α by the cocycle on (α, β) overlaps.*

Remark (relation to sheaf descent). In the discrete-poset setting with a finite target $\mathcal{T}$, Theorem TA-1 is the standard Čech-degree-1 descent statement: the cocycle condition on triples $\mathcal{C}^2$ maps to zero exactly when a section on the cover extends. The proof is executed here without cohomological language because the setting is finite and the target is a set alphabet. The theorem generalises to standard Borel targets by replacing the pointwise identity $T_{\alpha\beta} \circ M_\alpha = M_\beta$ with an almost-sure identity, at the cost of an almost-everywhere qualifier throughout.

3. Theorem TA-2: cocycle failure classifies obstructions

Setup (Theorem TA-2). Assume the presheaf $\mathcal{P}$ does *not* satisfy the cocycle: at least one triple (α, β, γ) has non-identity discrepancy $D_{\alpha\beta\gamma}$. We classify the shape of the failure by two quantities:

- **Rank.** $\text{rank}(D_{\alpha\beta\gamma}) := |\{a \in \mathcal{T} : D_{\alpha\beta\gamma}(a) \neq a\}|$ — the number of labels moved by the discrepancy permutation. Rank 0 iff $D_{\alpha\beta\gamma}$ is the identity.
- **Support.** The set of pairwise transitions that are non-trivial: $\text{supp}(\mathcal{P}) := \{(\alpha, \beta) : T_{\alpha\beta} \neq \text{id}_{\mathcal{T}}\}$. Encodes which overlaps contribute to the obstruction.

Theorem TA-2 (Cocycle failure classifies obstructions). *For a presheaf $\mathcal{P}$ on the context poset with at least three contexts and non-empty pairwise overlaps:*

1. (*glue.*) If $\text{rank}(D_{\alpha\beta\gamma}) = 0$ for every triple, $\mathcal{P}$ glues by Theorem TA-1 and no obstruction is present.
2. (*phase transition / boundary.*) If some $\text{rank}(D_{\alpha\beta\gamma}) \geq 1$ and $\text{supp}(\mathcal{P})$ is a strict subset of the pairwise overlap edges (i.e., at least one $T_{\alpha\beta} = \text{id}_{\mathcal{T}}$), the obstruction is **localised** to the non-trivial edges and admits a boundary interpretation: a region-change or scale/phase transition witnessed by a specific pair of contexts.
3. (*missing latent.*) If some $\text{rank}(D_{\alpha\beta\gamma}) \geq 1$ and every pairwise $T_{\alpha\beta}$ is non-identity, the obstruction is **spread across all overlaps** and admits a missing-latent interpretation: no chart is individually the anomalous one — the failure to glue is a collective property of the presheaf, consistent with the absence of a global label alphabet that would let all charts agree.

Proof.

(1) Immediate from Theorem TA-1: rank-zero discrepancy on every triple is the cocycle, so $\mathcal{P}$ glues.

(2) Suppose $T_{\alpha\beta} = \text{id}$ for some edge and some other triple has $\text{rank} \geq 1$. Then the presheaf restricted to the sub-poset of identity-edged pairs satisfies the cocycle on that sub-poset (compositions of identities are identities), so the failure is supported on $\text{supp}(\mathcal{P})$ — a proper subset of edges. Every gluing attempt through a chart in the trivial sub-poset agrees on that sub-poset; the disagreement is *only* along the non-trivial edges, and hence is diagnosable as a boundary between two

"self-consistent" regions joined by a non-trivial identification. This is the discrete analogue of a phase transition or scale change: one direction glues, the other does not.

(3) Suppose every $\tau_{\alpha\beta}$ is non-identity. Then no chart can be singled out as the anomalous one on the basis of $\text{supp}(\mathcal{P})$: every edge carries non-trivial identification, so removing any single chart still leaves a non-trivial residual presheaf. The failure to glue is a *joint* property of the transitions, not localised to a subset. The natural repair is to enlarge the target alphabet $\mathcal{T}$ — i.e., to posit an additional latent variable that all charts have been implicitly quotienting over — because in the enlarged $\mathcal{T}$ the transitions can be lifted to bijections that do close the cocycle. This is the operational content of "missing latent". A formal existence theorem for the enlargement — the smallest $\mathcal{T} \supseteq \mathcal{T}$ in which a lift closing the cocycle exists — is a separate result (a discrete analogue of universal covers of groupoids) that we do not prove here. $\square$

Corollary (category error as a rank saturation). *If $\text{rank}(D_{\alpha\beta\gamma}) = |\mathcal{T}|$ on every triple (the discrepancy has no fixed points), the presheaf carries no partial agreement between charts: every chart- α label is mapped to a different label by the loop $\alpha \rightarrow \beta \rightarrow \gamma \rightarrow \alpha$. This is the extreme missing-latent regime and, on finite target alphabets of small size, indistinguishable from a category error where the charts are measuring genuinely different things. On a two-element alphabet, for instance, $\text{rank} = 2$ is the only non-trivial possibility and the missing-latent and category-error signatures coincide; distinguishing them requires either a larger $\mathcal{T}$ or additional context beyond the presheaf. This is a limitation on the diagnostic resolution of Theorem TA-2, not on its statement.*

Corollary (relation to the parent paper's fiber audit). *A cocycle failure of missing-latent type witnesses a $\Delta_{\mathcal{Q}}(z) > 0$ regime in the fiber audit of parent-paper §5.4: the transitions identify chart-values across contexts as if they addressed the same z , but the loop discrepancy shows that the identification is not consistent — i.e., there is a hidden coordinate on the fibre that the charts have been marginalising over. Whether that hidden coordinate is task-relevant is a separate empirical question; the cocycle diagnostic identifies its existence, not its behaviour.*

4. Worked example: three contexts on the 4-bit Boolean world

World. $\mathbf{x} = \{0, 1\}^4$ (16 elements, uniform base distribution).

Contexts.

- $\mathbf{U}_1 := \{ \mathbf{x} : x_0 = 0 \}$ (8 worlds),
- $\mathbf{U}_2 := \{ \mathbf{x} : x_1 = 0 \}$ (8 worlds),
- $\mathbf{U}_3 := \{ \mathbf{x} : x_2 = 0 \}$ (8 worlds).

Pairwise overlaps: $\mathbf{U}_{12} = \{x_0 = x_1 = 0\}$ (4 worlds), $\mathbf{U}_{13} = \{x_0 = x_2 = 0\}$ (4 worlds), $\mathbf{U}_{23} = \{x_1 = x_2 = 0\}$ (4 worlds). Triple overlap $\mathbf{U}_{123} = \{x_0 = x_1 = x_2 = 0\}$ (2 worlds). The union $\mathbf{U}_1 \cup \mathbf{U}_2 \cup \mathbf{U}_3$ covers 14 of the 16 worlds; the two uncovered worlds are $(1, 1, 1, 0)$ and $(1, 1, 1, 1)$.

Target alphabet. $\mathcal{T} = \mathbb{Z}/4 = \{0, 1, 2, 3\}$.

Underlying observable. $g(\mathbf{x}) := (2 \cdot x_2 + x_3) \bmod 4$, valued in $\mathcal{T}$, defined on all of $\mathbf{x}$.

Chart maps (identical between the good and bad families):

- $M_1(\mathbf{x}) := g(\mathbf{x})$ on $\mathbf{U}_1$ — identity presentation on chart 1;
- $M_2(\mathbf{x}) := (g(\mathbf{x}) + 1) \bmod 4$ on $\mathbf{U}_2$ — shift-by-1 presentation;
- $M_3(\mathbf{x}) := (g(\mathbf{x}) + 2) \bmod 4$ on $\mathbf{U}_3$ — shift-by-2 presentation.

Good chart family $((+1, +1, +2))$. Transitions:

- $T_{12}(a) = (a + 1) \bmod 4$,
- $T_{23}(a) = (a + 1) \bmod 4$,
- $T_{13}(a) = (a + 2) \bmod 4$.

Cocycle discrepancy on the single triple $(1, 2, 3)$:

$$\begin{aligned}
D_{123}(a) &= T_{13}^{-1}(T_{23}(T_{12}(a))) \\
&= T_{13}^{-1}((a+1)+1) \\
&= (a+2)-2 \\
&= a \quad \text{for every } a \in \mathcal{I}.
\end{aligned}$$

D_{123} is the identity permutation on $\mathcal{I}$, $\text{rank}(D_{123}) = 0$. By Theorem TA-1 the presheaf glues, and the pivot-through-chart-1 construction ($\psi_1 = \text{id}$, $\psi_2 = T_{12}^{-1} = -1$, $\psi_3 = T_{13}^{-1} = -2$) yields the global theory

$$M(x) = g(x) = (2 \cdot x_2 + x_3) \bmod 4 \quad \text{on } U_1 \cup U_2 \cup U_3.$$

The 14-world union is covered consistently (every world in a multi-chart overlap receives a single candidate value).

Bad chart family $((+1, +1, +3))$. Same chart maps M_i . Transitions:

- $T_{12}(a) = (a+1) \bmod 4$,
- $T_{23}(a) = (a+1) \bmod 4$,
- $T_{13}(a) = (a+3) \bmod 4$ — differs from the good case only in the chart-1-to-chart-3 identification (the two-step consistent value would be $+2$; the bad table asserts $+3$).

Cocycle discrepancy:

$$\begin{aligned}
D_{123}(a) &= T_{13}^{-1}(T_{23}(T_{12}(a))) \\
&= T_{13}^{-1}((a+1)+1) \\
&= (a+2)-3 \\
&= (a-1) \bmod 4 \\
&= (a+3) \bmod 4.
\end{aligned}$$

D_{123} is the shift-by-3 permutation on $\mathbb{Z}/4$; $\text{rank}(D_{123}) = 4$ (no fixed points). Every pairwise transition is non-identity (each is a non-zero shift), so $\text{supp}(\mathcal{P}_{\text{bad}}) = \{(1,2), (1,3), (2,3)\}$ — the full edge set. By Theorem TA-2 clause (3), the bad presheaf is classified as *missing latent*.

Concretely, no global theory M on the union exists: the pivot-through-chart-1 attempt ($\psi_1 = \text{id}$, $\psi_2 = T_{12}^{-1} = -1$, $\psi_3 = T_{13}^{-1} = -3$) assigns two distinct candidate values to every world in the pairwise overlaps involving both chart 2 and chart 3 (6 worlds), witnessing the failure to glue as a set-theoretic inconsistency in the presheaf.

Phase-boundary reference family $((+1, \text{id}, \text{id}))$. Same chart maps. Transitions:

- $T_{12}(a) = (a+1) \bmod 4$,
- $T_{23}(a) = \text{id}(a) = a$,
- $T_{13}(a) = \text{id}(a) = a$.

Cocycle discrepancy:

$$\begin{aligned}
D_{123}(a) &= T_{13}^{-1}(T_{23}(T_{12}(a))) \\
&= \text{id}(\text{id}((a+1))) \\
&= (a+1) \bmod 4.
\end{aligned}$$

$\text{rank}(D_{123}) = 4$, but $\text{supp}(\mathcal{P}_{\text{phase}}) = \{(1,2)\}$ — a *strict* subset of the edges. By Theorem TA-2 clause (2), the phase-boundary reference is classified as *phase transition / boundary*: the disagreement is localised to a single overlap and admits the interpretation of a boundary between chart 1's regime and the (chart 2, chart 3)-agreeing regime.

The three verdicts — *glue*, *missing latent*, *phase transition* — separate the good, bad, and reference families cleanly under identical rank/support machinery, exhibiting TA-2's taxonomy in one computation.

5. Instrument: `experiments/theory_atlas_pair`

Exact witness of Theorems TA-1 and TA-2 on the setup above.

- Enumerate the 16-world Boolean cube; construct the three contexts, three pairwise overlaps, and one triple overlap by set-theoretic intersection.
- Represent each τ_{ij} as an explicit permutation of the 4-element target alphabet; permutation validity (bijection) is enforced by construction and re-checked by the test suite.
- For every triple (i, j, k) compute the cocycle discrepancy $\tau_{ik}^{-1} \circ \tau_{jk} \circ \tau_{ij}$ as a permutation on $\mathcal{X}$; report its rank (number of moved elements) and whether it equals the identity.
- Attempt to construct a global theory by pivoting through chart 1 ($\psi_1 = \text{id}$, $\psi_i = \tau_{1i}^{-1}$ for $i > 1$); flag every world in any pairwise overlap that receives more than one candidate value.
- Compute the taxonomy verdict per family (*glue* / *phase transition* / *missing latent*) from the rank/support rule of Theorem TA-2.

Pre-registered gates (all four pass exactly):

- `ta1_good_charts_satisfy_cocycle`: cocycle holds on every triple for the good family (discrepancy = identity).
- `ta1_bad_charts_violate_cocycle`: at least one triple has non-zero discrepancy for the bad family (rank ≥ 1).
- `ta1_glue_iff_cocycle`: pivot-through-chart-1 gluing is consistent for the good family and inconsistent for the bad family. Together with the previous two gates, this is the computational instantiation of Theorem TA-1's equivalence on this world.
- `ta2_bad_discrepancy_matches_missing_latent_signature`: bad family is classified as `missing_latent`; the phase-boundary reference is classified as `phase_transition`; the good family is classified as `glue`. Verifies Theorem TA-2's rank/support taxonomy on one presheaf-per-verdict.

The instrument is deterministic (no random seeds, no Monte Carlo); verdicts are exact. Bad-family discrepancy has rank 4 on $\mathbb{Z}/4$ (no fixed points), all three transitions are non-identity, gluing fails on exactly 6 of the 14 union worlds. Phase-boundary reference has rank 4 on the loop but exactly one non-identity transition and two identity transitions.

6. Relation to the SIC framework

Theorem TA-1 turns extended-program §5.5 from a *research direction* into a *theorem* about when a family of local theories can be integrated into a single global theory: exactly when the cocycle on triples holds. Theorem TA-2 turns the *failure* of the cocycle into a *diagnostic*: the rank / support signature of the discrepancy tells you what kind of disagreement you are looking at (localised boundary versus distributed missing-latent) — an informative obstruction rather than a bare inconsistency. Together with:

- Theorem CG-1 / CG-2 (concern as fiber geometry, Fisher metric and holonomy — `papers/concern_as_fiber_geometry/paper.md`),
- Theorem CT-1 / CT-2 (compiler tomography, MDL identifiability and compiler ecology — `papers/compiler_tomography/paper.md`),
- Theorem SA-1 (antecedent taxonomy — four canonical (U, P) recipes populating Theorem 4 — `papers/sufficient_antecedents/paper.md`),
- Theorem AF-1 / AF-2 (abstraction frontier as a Pareto antichain — `papers/abstraction_frontier/paper.md`),
- Theorem AG-1 / AG-2 (viability under bounded leakage — `papers/alignment_as_ensemble_governance/paper.md`),

the SIC extended program now has *seven* explicit theorem-instrument pairs beyond the six of the parent paper. The remaining §5 constructs (conditional rate–distortion control limit §5.2, causal semantics §5.7, representation-repair calculus §5.8, autocatalytic artwork §5.10) remain open — each is a direction rather than a theorem.

The sheaf framing sharpens the parent paper's honesty condition on theory integration: two disagreeing charts on overlapping contexts do not have to be reconciled into one — the honest question is whether their transitions close the cocycle, and, if not, whether the discrepancy signature indicates a boundary to respect or a missing latent to lift. The parent paper's Wigner-effectiveness worry (§5.5) becomes tractable: cross-domain theories glue exactly when the cocycle closes, and the rank / support of a failure reads off whether the "unreasonable" resemblance was a coincidence, a boundary, or a common latent waiting to be found.

7. Limitations

- **Three contexts and one triple.** The instrument uses three contexts, which yields exactly one triple and one cocycle discrepancy to inspect. The taxonomy is exercised on *edges* — the three pairwise transitions — so the missing-latent-vs-phase-boundary distinction is fully witnessed on this world. A follow-up instrument with four or more contexts (yielding multiple triples) would exhibit the rank/support classification *across* triples, at which point "spread across all overlaps" becomes literal on more than one axis.
 - **Finite target alphabet.** $\mathcal{T} = \mathbb{Z}/4$ is a small finite set; the proofs generalise to standard Borel targets with the pointwise cocycle identity replaced by almost-sure identities. The cohomological framing (Čech-degree-1 with values in the automorphism sheaf of $\mathcal{T}$) is the standard higher generalisation; it is not needed here and would not sharpen the finite-case statement.
 - **Category-error and missing-latent coincide at rank saturation.** On small alphabets the extreme missing-latent regime and a category error are indistinguishable from the presheaf alone — the corollary in §3 makes this explicit. Distinguishing them requires either a larger target or additional side information (e.g. a semantic signature on the labels).
 - **The lift theorem is not proved.** Theorem TA-2's missing-latent clause asserts the *natural repair* is to enlarge $\mathcal{T}$ so the cocycle can close; the formal existence of a smallest such lift (analogue of a universal cover of a groupoid) is a separate result that we do not prove. The instrument witnesses the diagnostic, not the repair.
 - **No Lean formalisation.** Both proofs are elementary and the sheaf descent argument in the discrete setting is amenable to Lean/mathlib once the permutation-composition machinery is imported. The formalisation is left for future work; the numerical witness in Instrument 5 provides the analogue check.
-

8. Reproduction

```
python3 experiments/theory_atlas_pair/experiment.py
python3 -m unittest tests.test_theory_atlas_pair
```

Full development is in the parent paper's §5.5 and the notes file `notes/structural_intelligence_conjecture.md`.